



ESD compliance at manufacturing facility

3D flexural capability is yet to catch up. Metal-clad PCBs manufactured in India are widely being used for LED applications helping meet mechanical design constraints. PCBs by Indian OEMs with various embedded components are under prototype qualification stages. Especially for consumer electronics, these are set to see wide usage in the next few quarters.

Though there has been commendable progress within PCB manufacturing, this segment is still at a nascent stage in India. The upward trend still stays within very few Indian and global PCB fabrication houses who have their manufacturing in India. Most of the PCB fabrication houses still lag behind and need to have the understanding of the international standards and customer expectations. This is a major reason for increased cost of the boards fabricated in India, resulting in high dependency on PCBs from abroad.

Solder and associated material

SMT is driving the advances in solder materials, as the requirements for fine-pitch printing and reliable solder joint with minimal solder volume become

a challenge. Smelting houses in India have been the main source of raw solder materials within the country for a long time.

With leading global solder consumable manufacturers sourcing raw materials from India, the manufacturing processes for smelting houses now meet world-class levels. Also, a few global solder consumable manufacturers have set up offices in India offering high-quality support, which, in turn, has raised the support requirements from Indian solder consumable manufacturers. This finally benefits the customers, specifically the CEMs who normally interact with more than one supplier.

Halogen-free requirement is another upcoming area of work. With this as a regulation, the challenges faced by the CEMs in India are very similar to the ones faced globally. This information is being used constructively by the solder material manufacturers.

Flux is now available in various forms—liquid, paste or solid—and various corrosion levels. This has significantly helped in soldering components of various types, which have various flux requirements within the same board, or which require more than one

type of solder. One good example is customised components with castellated and thermal pad on the bottom. In this case, flux in paste form is used as part of the solder paste and additional solid flux is used in castellated.

PCB surface finishes and the chemicals used also have a heavy influence on the direction in which SMT moves. OEMs and subsequently the PCB fabricators are moving away from the metallic (hot-air solder leveling) and electroplated finishes (electroless nickel immersion gold and electroless nickel electroless palladium immersion gold) towards organic finishes (organic solderability preservatives (OSP)s).

Major challenges with the organic surface finish are the fidelity of contact during the test of the PCB assembly and traces of OSP left behind. Advances in this area with conductive solder finishes (which is under qualification with a few Indian PCB fabricators) will help mitigate this challenge and streamline the use of organic surface finishes.

Various research projects for SMT are undertaken by Indian academic institutions in partnership with the industry. The primary focus is on alternative lead-free solder alloys with soldering profile similar to legacy Sn63/Pb37 (tin-lead) alloy and also alternative to the widely used SAC (An-Ag-Cu) alloys. PCB surface-finish defects elimination in the wet process is another primary research area.

Equipment

Majority of the equipment used for electronics manufacturing, cleaning and inspection are imported. All the major equipment manufacturers either have direct presence or authorised dealers. Majority of the equipment vendors excel in post-installation and annual maintenance support. With most high-speed electronics equipment and advanced soldering equipment available, advanced manufacturing using six-sigma approach and meeting stringent goals is being done in India.